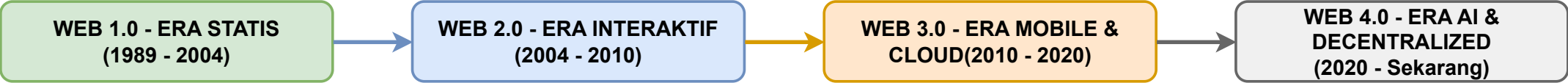


EVOLUSI TEKNOLOGI WEB (1989 - 2024)



- 1989-1991 - KELAHIRAN WEB
- Tim Berners-Lee menciptakan WWW
 - HTML pertama
 - HTTP protocol
 - URL/URI system
 - Web browser pertama

- 1993-1995 - WEB UNTUK PUBLIK
- Mosaic browser
 - HTML 2.0
 - CGI (Common Gateway Interface)
 - Netscape Navigator
 - SSL/TLS security

- 1995-2000 - WEB DINAMIS
- JavaScript (1995)
 - PHP (1995)
 - CSS (1996)
 - Flash (1996)
 - XML (1998)
 - Internet Explorer dominance

- 2000-2004 - ERA STANDARISASI
- XHTML 1.0
 - CSS 2.0
 - DOM Level 2
 - Web Standards Project
 - Mozilla Firefox (2004)

- 2004-2006 - REVOLUSI AJAX
- AJAX (Asynchronous JavaScript)
 - Gmail (2004)
 - Google Maps (2005)
 - jQuery (2006)
 - Social Media boom

- 2006-2008 - RICH INTERNET APPS
- Web 2.0 term popularized
 - User-generated content
 - Mashups & APIs
 - Adobe AIR
 - Microsoft Silverlight

- 2008-2010 - BROWSER WARS 2.0
- Google Chrome (2008)
 - HTML5 development starts
 - CSS3 modules
 - V8 JavaScript engine
 - WebKit proliferation

- 2009-2010 - MOBILE WEB MULAI
- iPhone & mobile browsers
 - Responsive web design concept
 - Node.js (2009)
 - AngularJS development
 - Cloud computing rise

- WEB 3.0 - ERA MOBILE & CLOUD
(2010 - 2020)

- 2010-2012 - HTML5 & STANDAR BARU
- HTML5 finalized
 - CSS3 widespread adoption
 - Responsive Web Design
 - Bootstrap framework
 - WebGL for 3D graphics

- 2012-2015 - JAVASCRIPT FRAMEWORKS
- AngularJS mainstream
 - React.js (2013)
 - Vue.js (2014)
 - Gulp/Grunt build tools
 - Single Page Applications

- 2015-2018 - MODERN WEB DEVELOPMENT
- ES6/ES2015 JavaScript
 - Webpack bundler
 - PWA (Progressive Web Apps)
 - GraphQL
 - Docker containerization

- 2018-2020 - FOKUS PERFORMA
- WebAssembly mainstream
 - JAMstack architecture
 - Static Site Generators
 - Core Web Vitals
 - HTTP/3 development

- WEB 4.0 - ERA AI & DECENTRALIZED
(2020 - Sekarang)

- 2020-2022 - FRAMEWORK MODERN
- Next.js 10+ with ISR
 - Svelte/SvelteKit popularity
 - Tailwind CSS mainstream
 - Micro-frontends
 - Web Components adoption

- 2022-2024- INTEGRASI AI
- ChatGPT & AI assistants
 - GitHub Copilot
 - AI-powered development
 - WebGPU for ML
 - Edge computing growth

- 2024 - SEKARANG - WEB TERDESENTRALISASI
- Web3 & Blockchain integration
 - Metaverse development
 - WebXR (AR/VR)
 - Quantum-resistant security
 - Green web technologies

- TREN MASA DEPAN
- Quantum computing integration
 - Brain-computer interfaces
 - Ambient computing
 - Sustainable web practices
 - Universal accessibility

TEKNOLOGI KUNCI BERDASARKAN ERA

- HTML • HTTP • CSS • JavaScript • PHP

- AJAX • jQuery • JSON • REST APIs • Flash

- HTML5 • React • Angular • Node.js • PWA

- AI/ML • WebAssembly • Web3 • WebXR • Quantum